27 June 1983

MILITARY SPECIFICATION

PROPELLANT M14 FOR USE IN
CARTRIDGE, 105MM TP-T, M490E1

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements, quality assurance provisions, and packaging of M14 Propellant for use in the Cartridge TP-T, M490E1 for the 105MM, M68 Gun.

1.2 Classification. The propellant shall be Type I Cylindrical Multi-Perforated grains.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.1) the following specifications and standards of issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitiation, form a part of this specification to the extend specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-286	-Propellants, Solid, Sampling Examination and Testing
MIL-STD-414	-Sampling Procedures and Tables for Inspection by Variables for percent defective
MIL-STD-652	-Propellants, Solid, for Cannons, Requirements and packing

2.1.2 Other government documents, drawings and publications. The following drawings form a part of this specification to the extent specified herein.

DRAWINGS (see 6.3)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9343010	-Case Cartridge and Liner Assembly 105MM M148A1B1
9329792	-Projectile, 105MM, TP-T, M489E1 Metal Parts Assembly
9343009	-Cartridge, 105mm, TP-T, M490E1

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The propellant shall be M14 Type I, conforming to the requirements of MIL-STD-652 except for the following:

3.1.1 Grain length to diameter ratio. The average grain length (L) shall be from 1.90 to 2.10 times the average grain diameter (D).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

3.1.2 Grain diameter to perforation diameter ratio. The average grain diameter (D) shall be from 11.0 to 14.0 times the average diameter of the perforation (d).

3.1.3 Web. The nominal web sizes shall be .050 inch (in.).

3.1.4 Bulk density. The nominal bulk density shall be 55.3 pounds per cubic foot (lb/ft^3) (See 4.5.2).

3.1.5 Total volatiles. The total volatiles content of the finished propellant shall be a maximum of 1.9% (see 4.5.3).

3.2 Ballistic assessment requirements. The propellant when fired in the M68 Gun with the M490E1 projectile shall comply with the following requirements at 70 degrees Fahrenheit (^{O}F) (see 4.5.1) and standard weapon conditions (see 6.6). The propellant shall be capable of assessment to a velocity of 3850 feet per second (fps) with an expected average pressure not to exceed 58,700 pounds per square inch (psi) and a charge weight not to exceed 199 ounces.

3.3 Ballistic uniformity requirements. The propellant when fired in the 68 Gun with the M490E1, projectile shall comply with the following requirements.

3.3.1 Uniformity of velocity. The standard deviation of velocity for the lot at 70°F shall not exceed 7.8 fps.

3.3.2 Average pressure. At 70°F and standard weapon conditions, the propellant shall not produce an average pressure greater than 58,700 psi. At 125°F and standard weapon conditions, the propellant shall not produce an average pressure greater than 68,600 psi.

3.3.3 Individual maximum pressure (IMP). The propellant shall not produce an individual pressure greater than 73,900 psi or an average pressure plus three (3) standard deviations greater than 71,500 psi at 125°F.

3.4 Preproduction inspection. This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples shall be as specified by the Contracting Officer.

3.5 Workmanship. The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. Preproduction Sample Inspection and Quality Conformance Inspection shall be conducted on this item.

4.3 Preproduction sample (pilot lots). The preproduction sample is intended to provide necessary ballistic information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. The Contractor shall select up to three propellant granulations which, from experience, are expected to bracket the requirements of this specification. The contractor shall submit preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of up to three pilot lots of a maximum of 3000 lb. each, representing the candidate granulations.

4.3.2 Inspections to be performed. Each of the granulations comprising the preproduction sample and submitted in accordance with 4.3.1. may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection - Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The contractor shall request approval from the Contracting Officer to produce the selected granulation. If none of the pilot lots submitted meets the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. All future production lots shall be of the same basic granulation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.4.2.1 Sampling for laboratory testing (see 3.1). Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical/physical properties in accordance with MIL-STD-652. (see 6.4 and 6.5).

4.4.2.2 Sampling for ballistic testing (see 3.2). Each lot of propellant shall be sampled for Proving Ground tests in accordance with MIL-STD-652.

4.4.2.3 Charge establishment (see 3.2). If the lot does not meet the applicable requirements, testing shall cease and the Test Director shall immediately notify the procuring activity and the technical agency.

4.4.2.4 Velocity uniformity (see 3.3.1). If the product of the factor 0.73 times the velocity standard deviation of the rounds tested in accordance with 4.5.1.2 exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.5 Average pressure (see 3.3.2). If the average pressure obtained exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.6 Individual Maximum Pressure (IMP).

4.4.2.6.1 If any individual pressure observed during the 125°F uniformity series, when corrected to standard conditions, recommended charge weight, and standard projectile weight exceeds the applicable requirement, the lot shall be rejected.

4.4.2.6.2 If the average pressure at 125°F plus three standard deviations exceeds the applicable requirement, apply referral criteria in accordance with 4.5.1.4. If the referral criteria cited do not apply, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

4.5 Test methods and procedures

4.5.1 Proving ground assessments. These tests shall be conducted at a Government Proving Ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. Tubes used for these assessments shall be restricted to fire only High Explosive Anti-Tank with Tracer (HEAT-T) ammunition. The wear of the tube shall be no greater than 0.015 inch when measured 25.25 inches from the rear of the tube. All test rounds shall be temperature conditioned in a large conditioning box, with adequate air circulation to assure uniform conditioning, at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all temperature conditionings: -50°F +5°F; 70°F ±2.5°F; 125°F -5°F and 145°F -5°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure. Test and calibration rounds shall be loaded and assembled by the Army Ammunition Plant LAP facility in accordance with the applicable drawing (see 4.5.4). .

4.5.1.1 Charge establishment (see Table 1).

4.5.1.1.1 Charge slope method. Three or more groups of rounds, three rounds for each group loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight), shall be conditoned and fired at 70°F. Three calibration rounds shall also be conditioned and fired at 70°F prior to the test rounds. From the data obtained, a composite charge weight versus velocity versus pressure curve shall be established. A charge weight shall be calculated which will yield the prescribed service velocity (see 3.2). This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.1.2.1). The difference between the corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established. The resultant will be the recommended charge weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

4.5.1.1.2 Constant slope method. After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite charge weight versus velocity versus pressure curve can be established from the test results generated from previous ten lots (see 4.5.1.1.1). This curve can then be used as in 4.5.1.1.1 to calculate the charge weight needed to achieve the service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method (see 4.5.1.2.1). The resultant will be the recommended charge weight.

4.5.1.2 Uniformity (see Table I).

4.5.1.2.1 Initial production. Forty (40) rounds shall be assembled with the calculated charge weight (see 4.5.1.1.1) of propellant. Ten rounds each shall be conditioned to -50°F, 70°F, 125°F, and 145°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at all temperatures shall be corrected to standard projectile weight (see 4.5.3) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall be calculated using corrected results. Testing at -50°F and 145°F shall be for information only. The initial production lots represent the first ten consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification.

4.5.1.2.2 Subsequent production. Twenty (20) rounds shall be assembled with the calculated charge weight (see 4.5.1.1.2) of propellant. Ten rounds each shall be conditioned to 70°F and 125°F (as indicated by thermocoupled control round). Rounds fired at 70°F shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration performance under test conditions and standard conditions). Firing results at both temperatures shall be corrected to standard projectile weight (see 4.5.3) and to recommended charge weight conditions (i.e., the difference between as fired charge weight and recommended charge weight). The means and standard deviations of the velocities and pressures at all firing temperatures shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

TABLE I. Proving Ground Test Summary

Test Phase	Temp	Sample Size		Velocity Requirements		Pressure Requirements		
	°F	Cal rd	Test rd	Service Velocity (fps)	Std Dev of Velocity (fps)		IMP (psi)	Average (psi)
CHG ESTAB	70	3	9	3850	INFO		INFO	58700
UNIF	-50*	N/A	10	INFO	INFO		INFO	INFO
	70	10	10	3850	7.8		INFO	58700
	125	N/A	10	INFO	INFO	**	73900	68600
	145*	N/A	10	INFO	INFO		INFO	INFO

* Performed for initial production testing only.

** The average pressure plus three (3) standard deviations shall not exceed 71500 (see 3.3.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

be calculated using corrected results. Throughout production, if 3 consecutive lots fail on first test, testing shall revert back to the initial production plan (see 4.5.1.2.1).

4.5.1.3 Retest criteria. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. The final adjustment in charge weight exceeds 2.0 oz of the charge weight used in uniformity firings, retest for uniformity in accordance with 4.5.1.2.

b. If for any reason, the Proving Ground considers that test conditions have detrimentally affected the test results, additional charges as required shall be tested.

4.5.1.4 Referral criteria. A test lot shall be referred to the procuring activity for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary.

a. Calibration rounds fired in any test phase fail to comply with Section 3 of this specification.

b. Calibration correction applied exceeds 25 fps.

c. Number of test or calibration results utilized in calculations is less than specified.

d. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, etc. Each such event shall be reported in detail on the applicable firing record.

e. Test lot fails velocity standard deviation or average pressure requirement at 70°F and any extreme test value is an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significance level of 0.05, the critical value (T_c) is 0.477. Calculate a test value (T_t) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{(X_9 - X_1)}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{(X_{10} - X_2)}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

Where:

X_1 = test result being tested as low outlier
X_2 = test result adjacent to X_1, after all the test results are ordered from smallest to largest
X_9 = test result adjacent to the largest test result
X_{10} = test result being tested as high outlier

X_1 or X_{10} is an outlier if T_t is greater than T_C.

4.5.2 Bulk density. The bulk density shall be determined in accordance with Method 502.2 of MIL-STD-286 except that a sample size of approximately 0.25 cubic feet shall be used and the 1000 milliliter graduate cylinder shall be replaced with a M148A1B1 cartridge case Dwg. 8865257.

4.5.3 Total volatiles. The total volatiles constent shall be determined in accordance with Method 103.5 of MIL-STD-286.

4.5.4 Projectile types. Projectiles may be of the following configurations:

Item	Type Obturator	Cart. Dwg.	Proj. Dwg.	Proj. Weight
105mm, TP-T, M490E1	1-Piece	9343009	9329792	20.50 ± .02 lb.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be in accordance with MIL-STD-652, except that all propellant designated for use as reference shall be packed level A.

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-A-48078 with the following exceptions:

6.1.1 Acquisition requirements. Acquisition requirements of MIL-A-48078 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.

6.1.2 Contract data requirements.

One copy of all description sheets and firing records required by this specification shall be forwarded to:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR-R
DRDAR-LCA-G
Dover, NJ 07801

Commander
US Army Armament Materiel Readiness Command
ATTN: DRSAR-QAM-A
Rock Island, IL 61299

6.2 Proving ground test summary (see Table I).

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.4 Surveillance sample. Samples shall be forwarded to US Army Armament Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-LCE-MP for the propellant surveillance test program in accordance with MIL-STD-652.

6.5 Reference sample. Whenever a propellant lot is selected for reference use, a 5-lb sample shall be forwarded to US Army Armament Research and Development Command, Dover, NJ 07801, ATTN: DRDAR-LCA-G for use in evaluating the propellant stockpile.

6.6 Standard weapon conditions. Standard weapon conditions refer to the weapon model and conditions of the test at the time the calibration series was fired.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1376-A189

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:02Z
Check the source to verify that this is the current version before use.